TWO GIANT BOXES IN RANDOM ALLOCATIONS

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ABSTRACT. Let $(Y_1, \dots, Y_n)$ be the random allocation of m balls into n boxes in which an allocation $(y_1, \dots, y_n)$ has probability proportional to $\prod_i w_{y_i}$, for a weight sequence $(w_k)_{k\ge 0}$. When $m/n \to \lambda > \nu$, where ν is determined by the weights, a positive fraction of the balls condenses into boxes whose sizes tend to infinity. For weights $w_k \sim ck^{-\beta}$ with $\beta > 2$ the largest box then contains $(\lambda - \nu)n + o_p(n)$ balls and the second largest $o_p(n)$. Janson asked whether the same holds for every nonincreasing weight sequence, and for every weight sequence with $w_{k+1}/w_k \to \infty$. We show that neither condition suffices. We construct nonincreasing weights for which, for every $\lambda > \nu$ and along a sequence of pairs (m, n) with $m/n \to \lambda$, each of the two largest boxes contains at least $(\lambda-\nu)n/13$ balls with probability tending to one, and weights with $w_{k+1}/w_k \ge e^k$ for which the second largest box contains more than $m/8$ balls with probability at least $1 - \exp(-m^3/256)$ whenever $m \ge 64$ is a power of two, $n \ge 2$ and $\log(m+n) \le m^2/256$. Both constructions give simply generated random trees in which, for infinitely many n, two nodes have outdegree of order n with probability tending to one.

1. INTRODUCTION

Let $w = (w_k)_{k\ge 0}$ be a sequence of nonnegative weights with $w_0 > 0$ and $w_k > 0$ for some $k \ge 1$. For integers $m \ge 0$ and $n \ge 1$ let $\mathcal{B}_{m,n}$ be the set of $y = (y_1, \dots, y_n) \in \mathbb{Z}^n_{\ge 0}$ with $y_1 + \cdots + y_n = m$, let $w(y) = \prod_{i=1}^n w_{y_i}$, and let $Z(m, n) = \sum_{y\in\mathcal{B}_{m,n}} w(y)$. When $Z(m, n) > 0$, the random allocation $B_{m,n} = (Y_1, \dots, Y_n)$ is the random element of $\mathcal{B}_{m,n}$ with

$$\mathbb{P}(B_{m,n} = y) = \frac{w(y)}{Z(m, n)}, \qquad y \in \mathcal{B}_{m,n}.$$

We think of Y_i as the number of balls in box i, and we write $Y_{(1)} \ge Y_{(2)} \ge \cdots \ge Y_{(n)}$ for $Y_1, \dots, Y_n$ in decreasing order. We follow the notation of Janson's survey [2], which treats random allocations together with simply generated random trees. If $\mathcal{T}_n$ denotes the simply generated random tree with n nodes and weights w, then by [2, Lemma 17.1] the outdegrees of the nodes of $\mathcal{T}_n$, taken as a multiset, have the same distribution as $Y_1, \dots, Y_n$ for $m = n - 1$.

Let $\Phi(t) = \sum_k w_k t^k$, let $\rho \in [0, \infty]$ be its radius of convergence, and let $\nu = \lim_{t\uparrow\rho} t\Phi'(t)/\Phi(t) \in (0, \infty]$ if $\rho > 0$, and $\nu = 0$ if $\rho = 0$. Suppose that $n \to \infty$ and $m = m(n)$ with $m/n \to \lambda$. If $\nu < \lambda < \infty$ and $w_k > 0$ for infinitely many k, then for suitable $K_n \to \infty$ the boxes with more than K_n balls contain $(\lambda - \nu)n + o_p(n)$ balls in total [2, Lemma 19.32]. The simplest way in which this condensation can happen is that the excess goes into a single box:

$$Y_{(1)} = (\lambda - \nu)n + o_p(n), \tag{1.1}$$

$$Y_{(2)} = o_p(n). \tag{1.2}$$

Jonsson and Stefánsson [4] showed this for simply generated trees with weights $w_k \sim ck^{-\beta}$, $\beta > 2$, and Janson [2, Theorem 19.34] proved (1.1)–(1.2) for random allocations with such weights and every $\lambda > \nu$. Janson also gave examples, with weights supported on sparse sets of

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integers, in which the excess is shared by two or more boxes along suitable sequences of (m,n) [2, Examples 19.37 and 19.38], and wrote that such behaviour is expected only for quite irregular weight sequences. He asked the following two questions.[1]

Problem (Janson [2, Problem 19.42])**.** *Suppose that $w_k \geq w_{k+1}$ for all (large) k. Does this imply that* (1.1)–(1.2) *hold when $\lambda > \nu$?*

Problem (Janson [2, Problem 19.43])**.** *Suppose that $w_{k+1}/w_k \to \infty$ as $k \to \infty$. (Hence, $\rho = 0$ and $\nu = 0$.) Does this imply that* (1.1)–(1.2) *hold when $\lambda > \nu$?*

The answer to both questions is negative.

Theorem 1.1. *There is a weight sequence w with $w_0 = w_1 = 1$ and $w_{k+1}/w_k \geq e^k$ for all $k \geq 1$ such that the following holds. If $m \geq 64$ is a power of two and $n \geq 2$ satisfies $\log(m+n) \leq m^2/256$, then*

$$\mathbb{P}(Y_{(2)} \leq m/8) \leq \exp(-m^3/256).$$

For $\lambda > 0$ let $m_i = 2^{i+1}$ and $n_i = \lceil m_i/\lambda \rceil$. Then $m_i/n_i \to \lambda$, and by Theorem 1.1, with probability tending to one, $Y_{(2)} > m_i/8$ and therefore $Y_{(1)} < 7m_i/8$. Thus neither (1.1) nor (1.2) holds for $(m,n) = (m_i, n_i)$, for any $\lambda > 0 = \nu$. Setting $m(n) = \lfloor \lambda n \rfloor$ for $n \notin \{n_i\}$ gives a sequence $m = m(n)$ with $m/n \to \lambda$ along which (1.1) and (1.2) fail. For the weights of Theorem 1.1 the ratio w_{k+1}/w_k is exceptionally large when k is a power of two; the weights in [2, Example 19.38] are supported on $\{0\}$ and the powers of two, while here every w_k is positive and $w_{k+1}/w_k \to \infty$. By contrast, for $w_k = k!$ and $m = n - 1$ the largest box contains $n - O_p(1)$ balls [2, Example 19.36]; simply generated trees with superexponential weights such as $((k-1)!)^\alpha$ are studied in [3].

Theorem 1.2. *There is a weight sequence w with $w_0 = w_1 = 1$ and $w_k \geq w_{k+1}$ for all $k \geq 0$, for which $\rho = 1$ and $0 < \nu < 0.7$, such that the following holds. For every $\lambda > \nu$ there are integers $n_1 < n_2 < \dots$ and $m_1, m_2, \dots$ with $m_r/n_r \to \lambda$ such that for every $\varepsilon > 0$, with $(m,n) = (m_r, n_r)$,*

$$\mathbb{P}\Big(Y_{(2)} \geq (\tfrac{1}{12} - \varepsilon)(\lambda - \nu)n \quad \text{and} \quad Y_{(1)} \leq (\tfrac{11}{12} + \varepsilon)(\lambda - \nu)n\Big) \to 1 \qquad (r \to \infty).$$

In particular neither (1.1) nor (1.2) holds along (m_r, n_r), and, as above, the pairs (m_r, n_r) can be completed to a sequence $m = m(n)$ with $m/n \to \lambda$. The weights of Theorem 1.2 satisfy $e^{-4}k^{-2\alpha} \leq w_k \leq e^3 k^{-\alpha}$ for $k \geq 1$, where $\alpha = 3/\log 2$ (Lemma 3.1). They drop sharply on a sparse sequence of dyadic blocks of box sizes. When the excess is comparable to such a block, it is cheaper to share it between two boxes below the block than to put it into one box above it. The drop acts as a cut-off on the box sizes, as in the zero-range processes with hop rates $u(k) = 1 + b/k + e^{k-aL}$ studied by Schwarzkopf, Evans and Mukamel [5]. There the stationary distribution on L sites is the random allocation with the weights $\prod_{j=1}^{k} u(j)^{-1}$, which are nonincreasing in k but depend on L, and they find, by a grand canonical analysis and numerical simulations, that the excess splits into several condensates of size at most aL. In Theorem 1.2 the weight sequence does not depend on n; the cut-offs at all the scales are part of one sequence.

For the trees, $m = n - 1$ and $\lambda = 1$.

Corollary 1.3. (i) *For the weights of Theorem 1.1 and $n = 2^{i+1}+1$ with $i \geq 5$, with probability at least $1 - \exp(-(n-1)^3/256)$ the tree $\mathcal{T}_n$ has two nodes of outdegree larger than $(n-1)/8$.*

(ii) *For the weights of Theorem 1.2 there is an infinite set of n along which, with probability tending to one, $\mathcal{T}_n$ has two nodes of outdegree at least $(1-\nu)n/13$ and no node of outdegree larger than $12(1-\nu)n/13$.*

[1] In the arXiv version of [2] they are Problems 18.41 and 18.42.

Theorem 1.1 is proved in Section 2. The weights of Theorem 1.2 are constructed in Section 3, the theorem is proved in Section 4, and Corollary 1.3 is proved in Section 5. Section 6 contains remarks, and Section 7 describes computer checks of the lemmas.

2. Proof of Theorem 1.1

Let $d_0 = 0$, let $d_j = j^3$ if $j \ge 1$ is a power of two and $d_j = j$ otherwise, and let

$$L(k) = \sum_{j=0}^{k-1} d_j, \qquad w_k = e^{L(k)} \qquad (k \ge 0).$$

Then $w_0 = w_1 = 1$, and $w_{k+1}/w_k = e^{d_k} \ge e^k$ for $k \ge 1$. Write $L(y) = Q(y) + S(y)$, where $Q(y)$ is the sum of the $j < y$ that are not powers of two and $S(y)$ is the sum of j^3 over the powers of two $j < y$.

Lemma 2.1. *For $y \ge 1$ we have $\frac{1}{2}y(y-1) - 2y < Q(y) \le \frac{1}{2}y(y-1)$. Moreover $S(0) = S(1) = 0$, and if $2^l < y \le 2^{l+1}$ with $l \ge 0$, then $S(y) = (8^{l+1}-1)/7$ and $S(y) < \frac{8}{7}\,4^l y$.*

Proof. If $2^l < y \le 2^{l+1}$, the powers of two below y are $1, 2, \dots, 2^l$; their sum is $2^{l+1} - 1 < 2y$ and the sum of their cubes is $\sum_{t=0}^{l} 8^t = (8^{l+1}-1)/7 < \frac{8}{7}\,8^l < \frac{8}{7}\,4^l y$. Since $Q(y)$ is $\frac{1}{2}y(y-1)$ minus the sum of the powers of two below y, the bounds on Q follow; for $y \le 1$ there are no powers of two below y. □

Lemma 2.2. *Let $i \ge 5$, $m = 2^{i+1}$, $a^* = 2^i + 1$, $b^* = 2^i - 1$ and $L^* = L(a^*) + L(b^*)$. If $y_1, \dots, y_n \ge 0$ satisfy $\sum_k y_k = m$ and at most one y_k exceeds $m/8$, then*

$$\sum_k L(y_k) \le L^* - \frac{m^3}{128}.$$

Proof. Let $D = m/8 = 2^{i-2}$, let $Y = \max_k y_k$, and fix k_0 with $y_{k_0} = Y$. All y_k with $k \ne k_0$ are at most D, and their sum is $m - Y$. If $2 \le y \le D$, then $2^l < y \le 2^{l+1}$ with $l \le i-3$, so $S(y) < \frac{8}{7}\,4^{i-3}y$ by Lemma 2.1; as $S(0) = S(1) = 0$,

$$\sum_{k \ne k_0} S(y_k) \le \tfrac{8}{7}\,4^{i-3}(m - Y).$$

By Lemma 2.1, $S(a^*) = (8^{i+1}-1)/7$ and $S(b^*) = (8^i - 1)/7$, so $S^* = S(a^*) + S(b^*) = (8^{i+1} + 8^i - 2)/7$.

If $Y \ge 2^i + 1$, then $m - Y \le 2^i - 1$ and $S(Y) = (8^{i+1}-1)/7$, since $2^i < Y \le 2^{i+1}$. Hence

$$\sum_k S(y_k) < \frac{8^{i+1}-1}{7} + \frac{8}{7}\,4^{i-3}2^i = \frac{8^{i+1}-1}{7} + \frac{8^{i-1}}{7},$$

and $S^* - \sum_k S(y_k) > (8^i - 1 - 8^{i-1})/7 = 8^{i-1} - \frac{1}{7}$. If $Y \le 2^i$, then $S(Y) \le (8^i - 1)/7$ and

$$\sum_k S(y_k) \le \frac{8^i - 1}{7} + \frac{8}{7}\,4^{i-3}2^{i+1} = \frac{8^i - 1}{7} + \frac{2 \cdot 8^{i-1}}{7},$$

so $S^* - \sum_k S(y_k) \ge (8^{i+1} - 1 - 2\cdot 8^{i-1})/7 > 8^{i-1}$. In both cases $S^* - \sum_k S(y_k) > 8^{i-1} - \frac{1}{7} = \frac{m^3}{64} - \frac{1}{7}$.

Next, $\sum_k \binom{y_k}{2} \le \binom{m}{2}$, so $\sum_k Q(y_k) \le \frac{1}{2}m(m-1)$ by Lemma 2.1. Since $a^* + b^* = m$ and $a^*(a^*-1) + b^*(b^*-1) = \frac{1}{2}m^2 - m + 2$, Lemma 2.1 also gives

$$Q(a^*) + Q(b^*) > \tfrac{1}{4}m^2 - \tfrac{1}{2}m + 1 - 2m = \tfrac{1}{4}m^2 - \tfrac{5}{2}m + 1,$$

and therefore $\sum_k Q(y_k) - Q(a^*) - Q(b^*) < \frac{1}{4}m^2 + 2m - 1$. Adding the two estimates,

$$\sum_k L(y_k) - L^* < \frac{m^2}{4} + 2m - \frac{6}{7} - \frac{m^3}{64} \le -\frac{m^3}{128},$$

because $m \ge 64$ gives $m^3/128 \ge m^2/2 \ge m^2/4 + 2m$. □

Proof of Theorem 1.1. Let $m = 2^{i+1}$ with $i \ge 5$, and let $n \ge 2$ with $\log(m+n) \le m^2/256$. Since $w_0 = 1$, the allocation with $y_1 = a^*$, $y_2 = b^*$ and $y_k = 0$ for $k \ge 3$ has weight e^{L^*}, so $Z(m,n) \ge e^{L^*}$. By Lemma 2.2, every allocation with $Y_{(2)} \le m/8$ has weight at most $e^{L^* - m^3/128}$, and there are $\binom{m+n-1}{m} \le (m+n)^m$ allocations. Hence

$$\mathbb{P}\big(Y_{(2)} \le m/8\big) \le (m+n)^m e^{-m^3/128} = \exp\Big(m\log(m+n) - \frac{m^3}{128}\Big) \le \exp\Big(-\frac{m^3}{256}\Big). \qquad \square$$

3. The weights of Theorem 1.2

Let $J_1 < J_2 < \cdots$ be positive integers with $J_{r+1} \ge 2J_r + 6$ for all r, for instance $J_r = 7\cdot 2^{r-1} - 6$, so that $(J_r) = (1, 8, 22, 50, \dots)$. We call the J_r cliffs. For integers $j \ge 0$ let

$$g(j) = \max\big(\{3j\} \cup \{6J_r + 4 : \ J_r \le j\}\big),$$

and let $w_0 = 1$ and $w_k = e^{-g(\lfloor \log_2 k \rfloor)}$ for $k \ge 1$. Let $\alpha = 3/\log 2$, so that $e^{-3j} = 2^{-\alpha j}$; note that $4.32 < \alpha < 4.33$. The distribution $\pi_k = w_k/\Phi(1)$ below is well defined by part (ii) of the following lemma.

Lemma 3.1. (i) *The function g is nondecreasing, $g(0) = 0$, and $3j \le g(j) \le 6j + 4$ for all j. Hence $w_0 = w_1 = 1$, $w_k \ge w_{k+1}$ for all $k \ge 0$, and $e^{-4}k^{-2\alpha} \le w_k < e^3 k^{-\alpha}$ for $k \ge 1$.*

(ii) *We have $\rho = 1$. The probability distribution $\pi_k = w_k/\Phi(1)$ on $\mathbb{Z}_{\ge 0}$ has mean $\nu = \Phi'(1)/\Phi(1)$ and finite variance σ^2, and $0 < \nu < \nu_0 < 0.7$, where $\nu_0 = 0.6243\dots$ is the value of ν for the weights $w_0^0 = 1$, $w_k^0 = e^{-3\lfloor \log_2 k \rfloor}$ $(k \ge 1)$.*

(iii) *If J is a cliff, then $g(j) = 3j$ for $\max(0, J-4) \le j \le J-1$, and $g(j) \ge 6J + 4$ for $j \ge J$.*

Proof. (i) The second set in the definition of g grows with j and consists of numbers at most $6j + 4$, so g is nondecreasing and $3j \le g(j) \le 6j + 4$; and $g(0) = 0$ because $J_1 \ge 1$. Hence $w_1 = e^{-g(0)} = 1 = w_0$ and (w_k) is nonincreasing. For $k \ge 1$ let $j = \lfloor \log_2 k \rfloor$, so that $k/2 < 2^j \le k$. Then $w_k \le e^{-3j} = 2^{-\alpha j} < (k/2)^{-\alpha} = e^3 k^{-\alpha}$, and $w_k \ge e^{-6j-4} = e^{-4}2^{-2\alpha j} \ge e^{-4}k^{-2\alpha}$.

(ii) By (i), $w_k^{1/k} \to 1$, so $\rho = 1$, and $\sum_k k^p w_k < \infty$ for $p < \alpha - 1$, in particular for $p \le 3$. Hence $\Phi(1)$, $\Phi'(1)$ and $\sum_k k^2 w_k$ are finite, π has finite variance, and $\nu = \lim_{t\uparrow 1} t\Phi'(t)/\Phi(t) = \Phi'(1)/\Phi(1)$, which is the mean of π. Since $w_1 > 0$, $\nu > 0$. Write Φ_0 for the generating function of w^0. Summing over the dyadic blocks $[2^j, 2^{j+1})$,

$$\Phi_0(1) = 1 + \sum_{j\ge 0} 2^j e^{-3j} = 1 + \frac{1}{1 - 2e^{-3}},$$

$$\Phi_0'(1) = \sum_{j\ge 0} e^{-3j} 2^{j-1}(3\cdot 2^j - 1) = \frac{3}{2(1-4e^{-3})} - \frac{1}{2(1-2e^{-3})},$$

which gives $\nu_0 = \Phi_0'(1)/\Phi_0(1) = 0.62433\dots$. Now $w_k \le w_k^0$ for all k, with equality for $k \le 1$, and $w \ne w^0$ because $g(J_1) = 6J_1 + 4 > 3J_1$. Let $\delta_k = w_k^0 - w_k$, $\Delta_0 = \sum_k \delta_k > 0$ and $\Delta_1 = \sum_k k\delta_k \ge 2\Delta_0$. Then

$$\nu = \frac{\Phi_0'(1) - \Delta_1}{\Phi_0(1) - \Delta_0} < \frac{\Phi_0'(1)}{\Phi_0(1)} = \nu_0,$$

since this inequality is equivalent to $\Delta_1/\Delta_0 > \nu_0$, and $\Delta_1/\Delta_0 \ge 2 > \nu_0$.

(iii) Let $J = J_r$. The second assertion holds by the definition of g. Let $\max(0, J-4) \le j \le J-1$. The cliffs $J_s \le j$ have $s < r$; if $r \ge 2$, each of them satisfies $6J_s + 4 \le 6J_{r-1} + 4 \le 3(J-6)+4 = 3J - 14 < 3j$. Hence $g(j) = 3j$. □

4. Proof of Theorem 1.2

In this section w is the weight sequence of Section 3, $\pi_k = w_k/\Phi(1)$, and $\xi, \xi_1, \xi_2, \dots$ are independent random variables with distribution π; let $S_k = \xi_1 + \dots + \xi_k$. Since $w(y) = \Phi(1)^n \prod_i \pi_{y_i}$, the allocation $B_{m,n}$ has the distribution of $(\xi_1, \dots, \xi_n)$ conditioned on $S_n = m$, and $\mathbb{P}(S_n = m) > 0$ because all w_k are positive.

We use the following truncated form of Bennett's inequality [1]; its proof is the usual one, applied to truncated variables.

Lemma 4.1. *Let $X_1, \dots, X_k$ be independent random variables with $\mathbb{E}X_i \le 0$ and $\sum_i \mathbb{E}X_i^2 \le V$, where $V > 0$. Then for $x > 0$ and $h > 0$,*

$$\mathbb{P}\Big(\sum_i X_i \ge x,\ \max_i X_i \le h\Big) \le \Big(\frac{eV}{V + xh}\Big)^{x/h}.$$

Proof. Let $Y_i = X_i \mathbf{1}\{X_i \le h\}$. On the event in question $\sum_i Y_i = \sum_i X_i$. Moreover $\mathbb{E}Y_i \le \mathbb{E}X_i \le 0$, $\mathbb{E}Y_i^2 \le \mathbb{E}X_i^2$ and $Y_i \le h$. Let $\psi(u) = e^u - 1 - u$. Since $\psi(u)/u^2 = \int_0^1 (1-s)e^{su}\,ds$ is nondecreasing in u, for $t > 0$ we have $\psi(tY_i) \le Y_i^2 \psi(th)/h^2$, and so

$$\mathbb{E}e^{tY_i} = 1 + t\mathbb{E}Y_i + \mathbb{E}\psi(tY_i) \le 1 + \frac{\psi(th)}{h^2}\mathbb{E}Y_i^2 \le \exp\Big(\frac{\psi(th)}{h^2}\mathbb{E}Y_i^2\Big).$$

By Markov's inequality and independence, $\mathbb{P}(\sum_i Y_i \ge x) \le \exp(V\psi(th)/h^2 - tx)$. Let $u = xh/V$ and $t = h^{-1}\log(1+u)$. Then $\psi(th) = u - \log(1+u)$, and

$$\frac{V\psi(th)}{h^2} - tx = \frac{x}{h} - \Big(\frac{x}{h} + \frac{V}{h^2}\Big)\log(1+u) \le \frac{x}{h}\big(1 - \log(1+u)\big),$$

which gives the claim. □

Let J be a cliff, let $N = 2^J$, $H = N/2$, $K = 3H$ and $D = N/8$, and let $n = n_J$ and $m = m_J$ be integers with

$$c_0 N \le n \le C_0 N, \qquad |m - \nu n - K| \le 1, \tag{4.1}$$

where $0 < c_0 \le C_0$ do not depend on J. In the rest of this section c and C denote positive constants that may depend on w, c_0 and C_0 but not on J, and whose values may change from one occurrence to the next; "for large J" means for all cliffs J larger than some constant. By Lemma 3.1(iii), the monotonicity of g and $g(j) \ge 3j$, and since $e^{-3J} = N^{-\alpha}$,

$$\pi_k = p_J := e^{-3(J-1)}/\Phi(1) \qquad \text{for } H \le k < N, \tag{4.2}$$

$$\pi_k \le e^{-3(J-3)}/\Phi(1) = e^9 N^{-\alpha}/\Phi(1) \qquad \text{for } D \le k < N, \tag{4.3}$$

$$\pi_k \le e^{-6J-4}/\Phi(1) = e^{-4}N^{-2\alpha}/\Phi(1) \qquad \text{for } k \ge N. \tag{4.4}$$

Lemma 4.2. *For large J, $\mathbb{P}(S_n = m) \ge cN^{3-2\alpha}$.*

Proof. For $1 \le i < j \le n$ let E_{ij} be the event that $\xi_i, \xi_j \in [H, N)$ and $\xi_l < H$ for all $l \ne i, j$; these events are disjoint. For an integer s let $\kappa(s)$ be the number of pairs $(a, b) \in [H, N)^2$ with $a + b = K + s$. A direct count gives $\kappa(s) = H - |s| - 1$ for $0 \le s < H$ and $\kappa(s) = H - |s| + 1$ for $-H < s < 0$, so $\kappa(s) \ge H - |s| - 1$ whenever $|s| < H$. By symmetry, independence and (4.2),

$$\mathbb{P}(S_n = m) \ge \sum_{i<j} \mathbb{P}(E_{ij},\, S_n = m) = \binom{n}{2} p_J^2 \sum_r \kappa(m - r - K)\, \mathbb{P}\Big(S_{n-2} = r,\ \max_{l \le n-2} \xi_l < H\Big).$$

Let $s_0 = 1 + 2\nu + 2\sigma\sqrt{n}$. If $|r - \nu(n-2)| \le 2\sigma\sqrt{n}$, then $|m - r - K| \le |m - \nu n - K| + 2\nu + |\nu(n-2) - r| \le s_0$, so

$$\mathbb{P}(S_n = m) \ge \binom{n}{2} p_J^2 \, (H - s_0 - 1) \, \mathbb{P}\Big(|S_{n-2} - \nu(n-2)| \le 2\sigma\sqrt{n}, \ \max_{l \le n-2} \xi_l < H\Big).$$

By Chebyshev's inequality, $|S_{n-2} - \nu(n-2)| > 2\sigma\sqrt{n-2}$ with probability at most $\frac{1}{4}$. By Lemma 3.1(i), $\mathbb{P}(\max_{l \le n-2} \xi_l \ge H) \le n \sum_{k \ge H} \pi_k \le CnH^{1-\alpha} \le CN^{2-\alpha} \to 0$, as $\alpha > 2$. Since $s_0 \le C\sqrt{N}$, for large J the last probability is at least $\frac{1}{2}$ and $H - s_0 - 1 \ge H/2$. Hence, using $n \ge c_0 N$ and $e^{-6(J-1)} = e^6 N^{-2\alpha}$,

$$\mathbb{P}(S_n = m) \ge \binom{n}{2} p_J^2 \, \frac{H}{4} \ge cN^2 \cdot N^{-2\alpha} \cdot N = cN^{3-2\alpha}. \qquad \square$$

Lemma 4.3. *For large J,*

$$\mathbb{P}\big(S_n = m \text{ and } \xi_i \ge D \text{ for at most one } i\big) \le C\big(N^{-11} + N^{1-2\alpha} + N^{-3-\alpha}\big).$$

Proof. Let $A_0 = \mathbb{P}(S_n = m, \ \max_i \xi_i < D)$ and let A_1 be the probability that $S_n = m$ and $\xi_i \ge D$ for exactly one i.

Let $X_i = \xi_i - \nu$, so that $\mathbb{E}X_i = 0$ and $\sum_{i \le n} \mathbb{E}X_i^2 = n\sigma^2$. If $\max_i \xi_i < D$, then $X_i \le D - 1 - \nu < D$ for all i, and if $S_n = m$, then $\sum_i X_i = m - \nu n \ge K - 1$. For $N \ge 8$ we have $(K-1)/D = 12 - 8/N \ge 11$ and $(K-1)D \ge N^2/8$. By Lemma 4.1 with $V = n\sigma^2$, $x = K - 1$ and $h = D$,

$$A_0 \le \Big(\frac{en\sigma^2}{(K-1)D}\Big)^{(K-1)/D} \le \Big(\frac{8eC_0\sigma^2}{N}\Big)^{11} \le CN^{-11},$$

where we used that the base is less than 1 for large J.

By symmetry,

$$A_1 \le n \sum_{k \ge D} \pi_k \, \mathbb{P}\Big(S_{n-1} = m - k, \ \max_{l \le n-1} \xi_l < D\Big).$$

By (4.4), and since the probabilities $\mathbb{P}(S_{n-1} = m - k)$ add up to at most 1, the terms with $k \ge N$ contribute at most $n\, e^{-4} N^{-2\alpha}/\Phi(1) \le CN^{1-2\alpha}$. For $D \le k < N$ we have $\pi_k \le e^9 N^{-\alpha}/\Phi(1)$ by (4.3), and

$$m - k - \nu(n-1) \ge m - \nu n - (N-1) + \nu \ge K - N + \nu \ge N/2.$$

The events $\{S_{n-1} = m - k\}$ are disjoint, so by Lemma 4.1 with $V = (n-1)\sigma^2$, $x = N/2$ and $h = D$, the terms with $D \le k < N$ contribute at most

$$n \, \frac{e^9 N^{-\alpha}}{\Phi(1)} \, \mathbb{P}\Big(\sum_{l \le n-1} (\xi_l - \nu) \ge \frac{N}{2}, \ \max_{l \le n-1} \xi_l < D\Big) \le CN^{1-\alpha}\Big(\frac{16en\sigma^2}{N^2}\Big)^4 \le CN^{-3-\alpha}.$$

Adding the three bounds proves the lemma. $\square$

Proposition 4.4. *Let J run through the cliffs and let $(m, n) = (m_J, n_J)$ satisfy* (4.1)*. Then $\mathbb{P}(Y_{(2)} \ge N/8) \to 1$, and $\mathbb{P}(Y_{(1)} \le 11N/8 + \varepsilon n) \to 1$ for every $\varepsilon > 0$.*

Proof. The event $Y_{(2)} < D$ is the event that at most one box contains D or more balls. By Lemmas 4.2 and 4.3,

$$\mathbb{P}(Y_{(2)} < D) \le \frac{C\big(N^{-11} + N^{1-2\alpha} + N^{-3-\alpha}\big)}{cN^{3-2\alpha}} = C\big(N^{2\alpha-14} + N^{-2} + N^{\alpha-6}\big) \to 0,$$

since $\alpha < 5$.

Let $\varepsilon > 0$. If $Y_{(1)}+Y_{(2)} > m-\nu n+\varepsilon n/2$, then there are boxes $i \neq j$ with $Y_i+Y_j > m-\nu n+\varepsilon n/2$, and the other $n-2$ boxes contain fewer than $\nu n - \varepsilon n/2$ balls. Hence

$$\mathbb{P}\big(Y_{(1)} + Y_{(2)} > m - \nu n + \varepsilon n/2\big) \le \binom{n}{2} \frac{\mathbb{P}(S_{n-2} < \nu n - \varepsilon n/2)}{\mathbb{P}(S_n = m)}.$$

For $\theta \ge 0$ and $x \ge 0$ we have $e^{-\theta x} \le 1 - \theta x + \theta^2 x^2/2$, so $\mathbb{E}e^{-\theta\xi} \le \exp(-\theta\nu + \theta^2\mathbb{E}\xi^2/2)$, and Markov's inequality gives $\mathbb{P}(S_k \le k\nu - t) \le \exp(-\theta t + k\theta^2\mathbb{E}\xi^2/2)$ for $t \ge 0$. With $k = n-2$, $t = \varepsilon n/2 - 2\nu \ge \varepsilon n/4$ and $\theta = t/(k\mathbb{E}\xi^2)$ this gives $\mathbb{P}(S_{n-2} < \nu n - \varepsilon n/2) \le \exp(-\varepsilon^2 n/(32\mathbb{E}\xi^2))$ for large n. Since $\mathbb{P}(S_n = m) \ge cN^{3-2\alpha}$ and $c_0 N \le n \le C_0 N$, the right-hand side above tends to 0. On the complementary event, if also $Y_{(2)} \ge D$, then

$$Y_{(1)} \le m - \nu n + \frac{\varepsilon n}{2} - D \le K + 1 + \frac{\varepsilon n}{2} - \frac{N}{8} = \frac{11N}{8} + 1 + \frac{\varepsilon n}{2} \le \frac{11N}{8} + \varepsilon n$$

for large J. □

Proof of Theorem 1.2. Take the cliffs $J_r = 7 \cdot 2^{r-1} - 6$ in Section 3; by Lemma 3.1 the weights have the stated properties. Let $\lambda > \nu$, and for a cliff J let $n_J = \lceil K/(\lambda-\nu) \rceil$ and $m_J = \lfloor \nu n_J + K \rfloor$. Then (4.1) holds for large J with $c_0 = 3/(2(\lambda-\nu))$ and $C_0 = 2/(\lambda-\nu)$, n_J is strictly increasing in J for large J, and $K/n_J \to \lambda - \nu$ and $m_J/n_J \to \lambda$. Since $N/8 = K/12$ and $11N/8 = 11K/12$, Proposition 4.4 gives, for every $\varepsilon > 0$, with probability tending to one,

$$Y_{(2)} \ge \frac{K}{12} \ge \Big(\frac{1}{12} - \varepsilon\Big)(\lambda-\nu)n \quad \text{and} \quad Y_{(1)} \le \frac{11K}{12} + \frac{\varepsilon}{2}(\lambda-\nu)n \le \Big(\frac{11}{12} + \varepsilon\Big)(\lambda-\nu)n$$

for large J. The pairs (m_J, n_J) for the cliffs J from the point on where n_J is strictly increasing are the pairs (m_r, n_r) of the theorem. □

5. Trees

Proof of Corollary 1.3. By [2, Lemma 17.1], the outdegrees of $\mathcal{T}_n$ have, as a multiset, the distribution of $Y_1, \dots, Y_n$ for $m = n-1$. (i) Let $n = 2^{i+1} + 1$ with $i \ge 5$, so that $m = n - 1 = 2^{i+1} \ge 64$ and $\log(m+n) = \log(2m+1) \le m^2/256$. Theorem 1.1 gives the claim. (ii) Let $n_J = \lceil (K+1)/(1-\nu) \rceil$ and $m_J = n_J - 1$ for the cliffs J; recall that $\nu < 1$. Then $(1-\nu)n_J \ge K+1$ and $(1-\nu)n_J < K + 2 - \nu$, so $K \le m_J - \nu n_J < K+1$, and (4.1) holds for large J. Since $K/n_J \to 1-\nu$, Proposition 4.4 gives, with probability tending to one, $Y_{(2)} \ge K/12 \ge (1-\nu)n_J/13$ and $Y_{(1)} \le 11K/12 + \varepsilon n_J \le 12(1-\nu)n_J/13$, when $\varepsilon < (1-\nu)/312$ and J is large. □

6. Remarks

Remark 6.1. In both constructions, replacing two large boxes by one larger box and one small box lowers the product of their weights, even when the small box holds a constant multiple of n balls. In Section 2, $w_{m-j}w_j < w_{a^*}w_{b^*}$ for $1 \le j \le m/4$: as in the proof of Lemma 2.2, $S(a^*) + S(b^*) - S(m-j) - S(j) \ge 8^{i-1}$ and $Q(m-j) + Q(j) - Q(a^*) - Q(b^*) < m^2/4 + 2m$. In Section 4, if $a, b \in [N/2, N)$ and $0 \le j \le a + b - N$, then $w_{a+b-j}w_j \le e^{-10}w_a w_b$ by (4.2) and (4.4). For the distribution π of Section 4 and k near $3N/2$, the ratio $\mathbb{P}(\xi_1 + \xi_2 = k)/\mathbb{P}(\xi = k)$ is of order N, whereas for weights $w_k \sim ck^{-\beta}$ with $\beta > 2$ it tends to 2 as $k \to \infty$.

Remark 6.2. The theorems concern particular sequences: in Theorem 1.1 the number of balls is a power of two, and in Theorem 1.2 the excess $m - \nu n$ lies within 1 of $3 \cdot 2^{J-1}$ for a cliff J. We do not determine the behaviour of $Y_{(1)}$ and $Y_{(2)}$ along other sequences, nor their joint limit law along (m_r, n_r). Nor do the constructions bear on Problems 19.40 and 19.41 of [2], which ask whether $Y_{(1)}/n \to 0$ in probability is possible when $0 < \nu < \lambda$ and when $\lambda > \nu = 0$.

7. Computer checks

The script `code/checks.py` in the repository listed under Data availability uses only the Python standard library, and its output is `code/checks.out`. Identities and inequalities between integers and rational numbers are checked exactly. The moments of π are computed in double precision by summing over the dyadic blocks, and probabilities are computed from the generating functions, without sampling, in double precision, by repeated squaring of power series truncated at degree m. The probabilities compared with bounds are sums of positive terms; for the weights of Section 2 they are computed from the logarithms of the terms.

For the weights of Section 2, the script checks $L = Q + S$ and Lemma 2.1 for every $y \le 4096$, and $w_{k+1}/w_k = e^{d_k}$ with $d_k \ge k$ for every $k < 4096$. For $2 \le i \le 12$ and $m = 2^{i+1}$ it computes, by dynamic programming over the box sizes, the exact maximum of $\sum_k L(y_k)$ over all $y_1, y_2, \ldots \ge 0$ with $\sum_k y_k = m$ and at most one y_k larger than $m/8$, and it checks the steps of the proof of Lemma 2.2 on the corresponding maxima of $\sum_k S(y_k)$ and $\sum_k Q(y_k)$. For $i \ge 5$ the ratio $(L^* - \max \sum_k L(y_k))/m^3$ lies between 0.0138 and 0.0163, against $1/128 < 0.0079$ in Lemma 2.2; it is 0.0096 for $i = 4$, 0.0007 for $i = 3$, and negative for $i = 2$. For $(m, n) = (64, 32)$, $(64, 64)$, $(64, 65)$, $(64, 128)$, $(128, 128)$, $(128, 129)$, $(128, 256)$ and $(256, 512)$ the script computes $\log \mathbb{P}(Y_{(2)} \le m/8)$, which lies between 2.32 and 2.41 times the bound $m \log(m + n) - m^3/128$ from the proof of Theorem 1.1; for $(m, n) = (64, 32)$ the two numbers are -4085.8 and -1755.9. As controls, for the weights $w_k = e^{k(k-1)/2}$, which also satisfy $w_{k+1}/w_k \to \infty$, the maximum exceeds L^* for every i, and $\mathbb{P}(Y_{(2)} \le m/8)$ equals 1 up to rounding error for $m \le 128$.

For the weights of Section 3, the script checks Lemma 3.1(i) and (iii) for $j \le 400$ and the cliff sequences with $1 \le J_1 \le 12$ and $J_{r+1} = 2J_r + c$, $c \in \{6, 7, 9\}$ (29,824 checks); with $J_{r+1} = 2J_r + 4$ the identity $g(J - 3) = 3(J - 3)$ fails at 58 cliffs. For $g(j) = 3j$ it obtains $\Phi_0(1) = 2.110586$ and $\nu_0 = 0.624335$, in agreement with the closed forms in the proof of Lemma 3.1, and for the sequences $J_{r+1} = 2J_r + 6$ with $1 \le J_1 \le 12$ it finds $0 < \nu < \nu_0$; for the sequence of Theorem 1.2, which has $J_1 = 1$, it obtains $\nu = 0.503957$. It checks the bound $\kappa(s) \ge H - |s| - 1$ from the proof of Lemma 4.2 for $3 \le J \le 14$ and all $|s| < H$ (32,748 checks). It checks Lemma 4.1 for $X_i = \xi_i - \nu$, where the ξ_i have the law π for the cliffs starting at $J_1 = 8$, with $k \in \{5, 10, 30, 100\}$, $h \in \{1, 2, 4, 8, 16, 32\}$ and $x/h \in \{1, 2, 4, 8, 12\}$: in these 120 cases the largest ratio of the probability to the bound is 0.0858, and without the event $\max_i X_i \le h$ the bound fails in 41 of them.

Finally, the script computes the law of $Y_{(2)}$ at the cliffs $J = 8, 9, 10$ for the cliff sequence that starts at J ($J_1 = J$ and $J_{r+1} = 2J_r + 6$), and at $J = 8$ for the sequence of Theorem 1.2. It takes (m, n) as in the proof of Theorem 1.2 with $\lambda - \nu = 1$, that is $n = K$ and $m = K + \lfloor \nu K \rfloor$, and as in the proof of Corollary 1.3(ii) for the trees. Table 1 gives $\mathbb{P}(Y_{(2)} < N/8)$. The control is $\mathbb{P}(Y_{(2)} \ge N/8)$ for the weights without cliffs, $g(j) = 3j$, at the pair (m, n) given by the same rule with ν_0 in place of ν; it is below 1% in every case. For $J_1 = 1$ the two rules give different pairs, since $\nu = 0.504$. At the pairs listed in the table for $J_1 = 1$, the excess $m - \nu_0 n$ of the weights without cliffs is about $1.32N$ for $\lambda - \nu = 1$ and $1.14N$ for the tree, instead of $1.5N$, and there $\mathbb{P}(Y_{(2)} \ge N/8)$ is $5.3 \cdot 10^{-3}$ and $2.7 \cdot 10^{-2}$. The probabilities that $Y_{(2)} < N/4$ are larger and decrease in the same way; for the trees they are 0.135, $7.5 \cdot 10^{-3}$ and $3.4 \cdot 10^{-4}$ for $J = 8, 9, 10$.

These computations test the identities and inequalities used in the proofs; they are not part of them.

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TABLE 1. $\mathbb{P}(Y_{(2)} < N/8)$ at a cliff J, $N = 2^J$, and the control $\mathbb{P}(Y_{(2)} \geq N/8)$ for the weights without cliffs.

		$\lambda - \nu = 1$				trees, $m = n - 1$		
J	J_1	n	m	$\mathbb{P}(Y_{(2)} < N/8)$	control	n	$\mathbb{P}(Y_{(2)} < N/8)$	control
8	8	384	623	$5.2 \cdot 10^{-5}$	$2.4 \cdot 10^{-3}$	1025	$3.5 \cdot 10^{-2}$	$7.1 \cdot 10^{-3}$
9	9	768	1247	$8.3 \cdot 10^{-8}$	$4.6 \cdot 10^{-4}$	2048	$1.8 \cdot 10^{-4}$	$1.3 \cdot 10^{-3}$
10	10	1536	2494	$2.1 \cdot 10^{-10}$	$9.1 \cdot 10^{-5}$	4092	$1.1 \cdot 10^{-7}$	$2.5 \cdot 10^{-4}$
8	1	384	577	$1.6 \cdot 10^{-6}$	$2.4 \cdot 10^{-3}$	777	$6.7 \cdot 10^{-5}$	$7.1 \cdot 10^{-3}$

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. Problems 19.42 and 19.43 of [2] were among the open problems in probability on which the project's agents worked in September 2026. On 13 and 14 September 2026 Claude Fable 5.1 (Anthropic) found the constructions of Theorems 1.1 and 1.2 and their proofs. On 12 and 13 October 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, wrote the proofs in their present form, searched the literature, and wrote the computer checks and the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code is available at
`https://github.com/yspennstate/condensation-two-giant-boxes`.

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